dimensional space, as in Figure 13.3b, but the electromagnetic force—eternally trapped within our dimensions—would be unable to reveal this.

Okay, you might say, but the electromagnetic force is only one of nature's four forces. What about the other three? Can they probe into the extra dimensions, thus enabling us to reveal their existence? For the strong and weak nuclear forces, the answer is, again, no. In the braneworld scenario, calculations show that the messenger particles for these forces—gluons and W and Z particles—also arise from open-string vibrational patterns, so they are just as trapped as photons, and processes involving the strong and weak nuclear forces are just as blind to the extra dimensions. The same goes for particles of matter. Electrons, quarks, and all other particle species also arise from the vibrations of open strings with trapped endpoints. *Thus, in the braneworld scenario, you and I and everything we've ever seen are permanently imprisoned within our three-brane.*Taking account of time, everything is trapped within our four-dimensional slice of spacetime.

Well, almost everything. For the force of gravity, the situation is different. Mathematical analyses of the braneworld scenario have shown that gravitons arise from the vibrational pattern of closed strings, much as they do in the previously discussed no-braner scenarios. And closed strings— strings with no endpoints—are not trapped by branes. They are as free to leave a brane as they are to roam on or through it. So, if we were living in a brane, we would not be completely cut off from the extra dimensions. Through the gravitational force, we could both influence and be influenced by the extra dimensions. Gravity, in such a scenario, would provide our sole means for interacting beyond our three space dimensions.

How big could the extra dimensions be before we'd become aware of them through the gravitational force? This is an interesting and critical question, so let's take a look.

Gravity and Large Extra Dimensions

Back in 1687, when Newton proposed his universal law of gravity, he was actually making a strong statement about the number of space